

Combinatorial Geometry

Problem 1. Given are 50 points in the plane, no three of them belonging to a same line. Each of these points is coloured using one of four given colours. Prove that there is a colour and at least 130 scalene triangles with vertices of that colour.

Problem 2. Prove that in any convex polygon where all pairwise distances between vertices are distinct, there exists a vertex such that the closest vertex of the polygon is adjacent to it.

Problem 3. There are $2m$ red points and $2n$ blue points on the plane such that no three points are collinear. Prove that there exists a line that separates all points into two halves, such that each half contains exactly m red points and n blue points.

Problem 4. There is a convex polygon $\mathcal{P}$ on the coordinate plane. It is known that no matter how we translate $\mathcal{P}$ on the plane (without rotating and reflecting it), there is always a lattice point (a point with integer coordinates) covered by $\mathcal{P}$. Prove that there are two points on or inside $\mathcal{P}$ whose distance is at least $\sqrt{2}$.

Problem 5. A square grid on the Euclidean plane consists of all points (m, n) , where m and n are integers. Is it possible to cover all grid points by an infinite family of discs with non-overlapping interiors if each disc in the family has radius at least 5?